

Size of stars

As long as stars are on the main sequence, their size stays constant.
Their luminosity depends only on their temperature.
Bigger stars → larger luminosity, larger temperature

But when they leave the main sequence (end of life), size can
Increase/ decrease drastically and a cooler but really big star
Can be more luminous than a hotter star. (red giant)

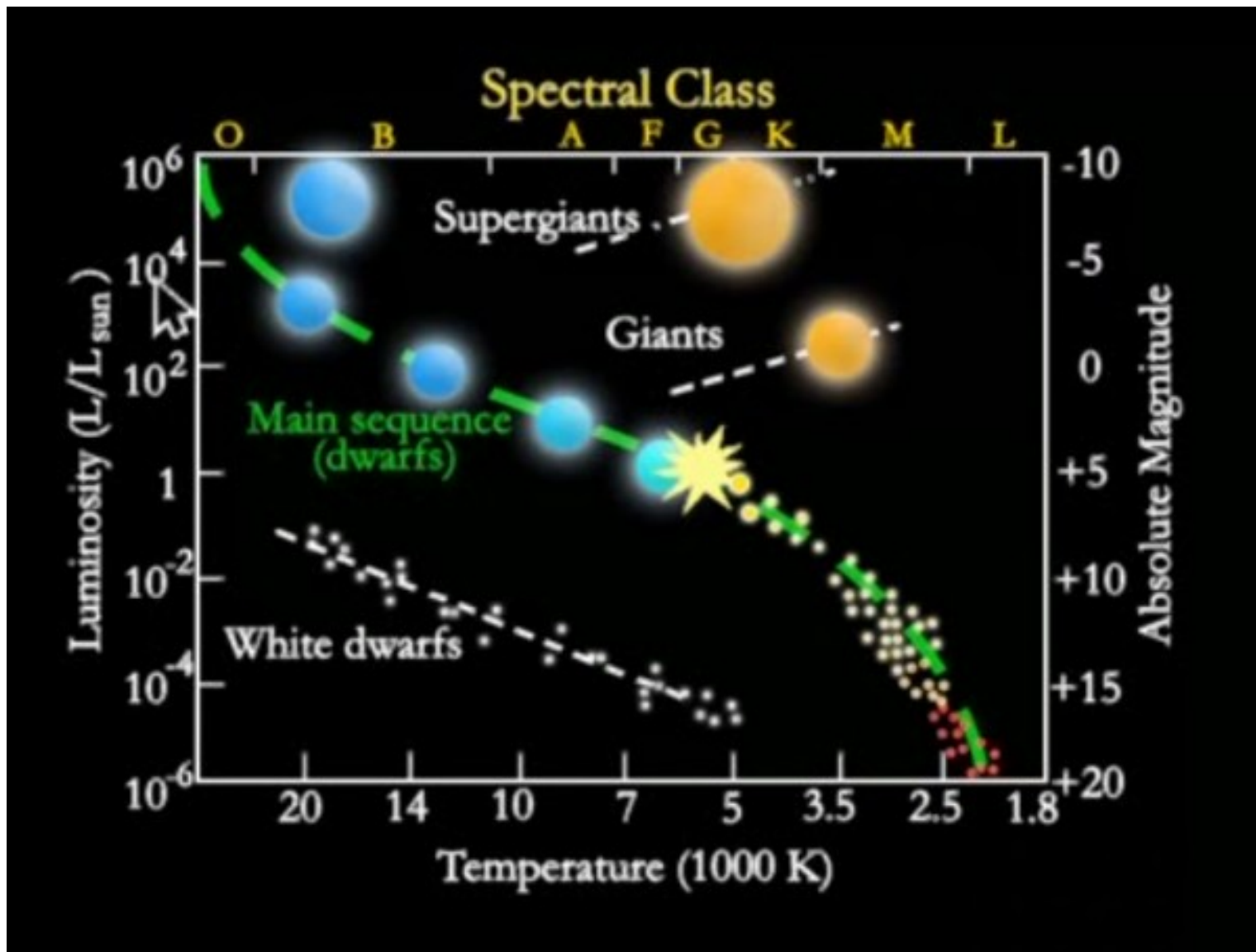
Or a hot but small star can be less luminous than a cooler star.
(white dwarf)

Luminosity depends on the surface area. Its proportional to R^2

For a sphere of radius R , Surface Area = $4\pi R^2$, so that

$$L = 4\pi R^2 \sigma T^4$$

Note: we can find the size
Of stars on main sequence
Using this relationship.
You know $T, L \rightarrow R$



Movie → range of sizes

Luminosity (or Power)

This is the amount of energy emitted per second by the **entire** object.

$$L = (\text{Surface Area}) \times \left(\frac{\text{energy emitted}}{\text{cm}^2 \text{ sec}} \right)$$

$$L = (\text{Surface Area}) \times \sigma T^4$$

For a sphere of radius R , Surface Area = $4\pi R^2$, so that

$$L = 4\pi R^2 \sigma T^4$$

If we know both L and T , then we can derive the radius, R !

SO IF you increase the size of the object, you increase its LUMINOSITY (brightness). So if a STAR is large enough, it can OUTPOWER a hotter but SMALLER star



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Consider a luminous (disco) Lamp.

Energy output of each square
Proportional to the temperature raised
To the 4th power →
the Stefan-Boltzmann law.



The energy produced by **each square** (unit area) of the lamp and per second = σT^4 if the lamps were to be a black body (thermal emitter).

. σ is a constant.

The energy produced by the whole lamp per second = **Luminosity** (sometimes called intrinsic Brightness) of the lamp (or power in watts) = energy output (from fusion) .

Luminosity = Multiply σT^4 by the area = total number of squares.

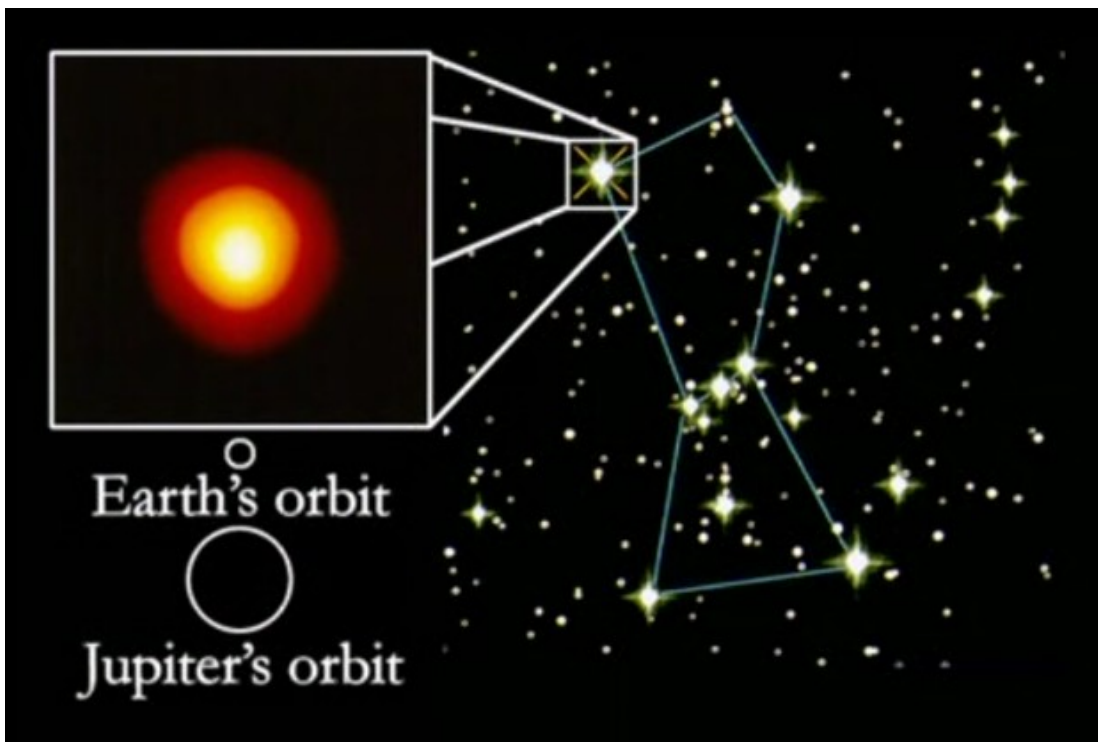
= (energy per square) x (total number of squares)

Luminosity = $\sigma T^4 \times 4\pi R^2$ $4\pi R^2$ is the area of a sphere of radius R



From your book (seeds/backman) lava is as hot as a candle but more luminous because More surface area. You need to wear gears to protect your eyes.

So power (energy produced per unit second) also depends on the surface area

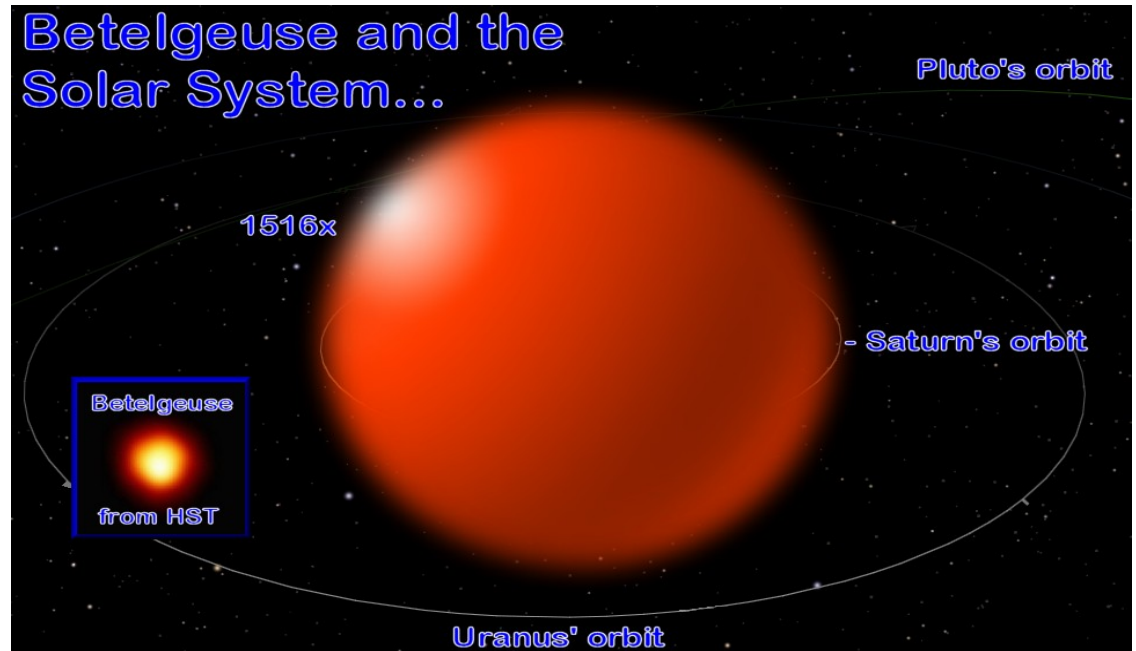


For a sphere of radius R , Surface Area = $4\pi R^2$, so that

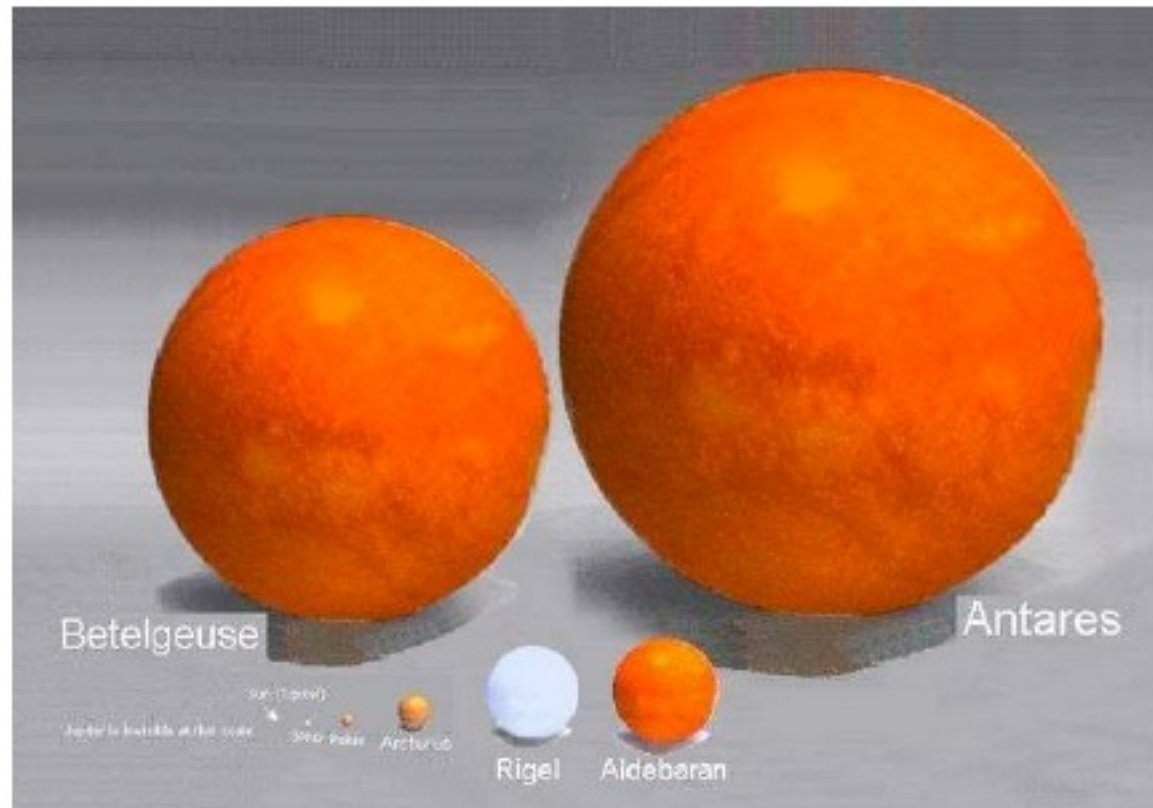
$$L = 4\pi R^2 \sigma T^4$$

For the same temperature,
If the radius x 1000
The luminosity is x 1000,000

Betelgeuse and the Solar System...

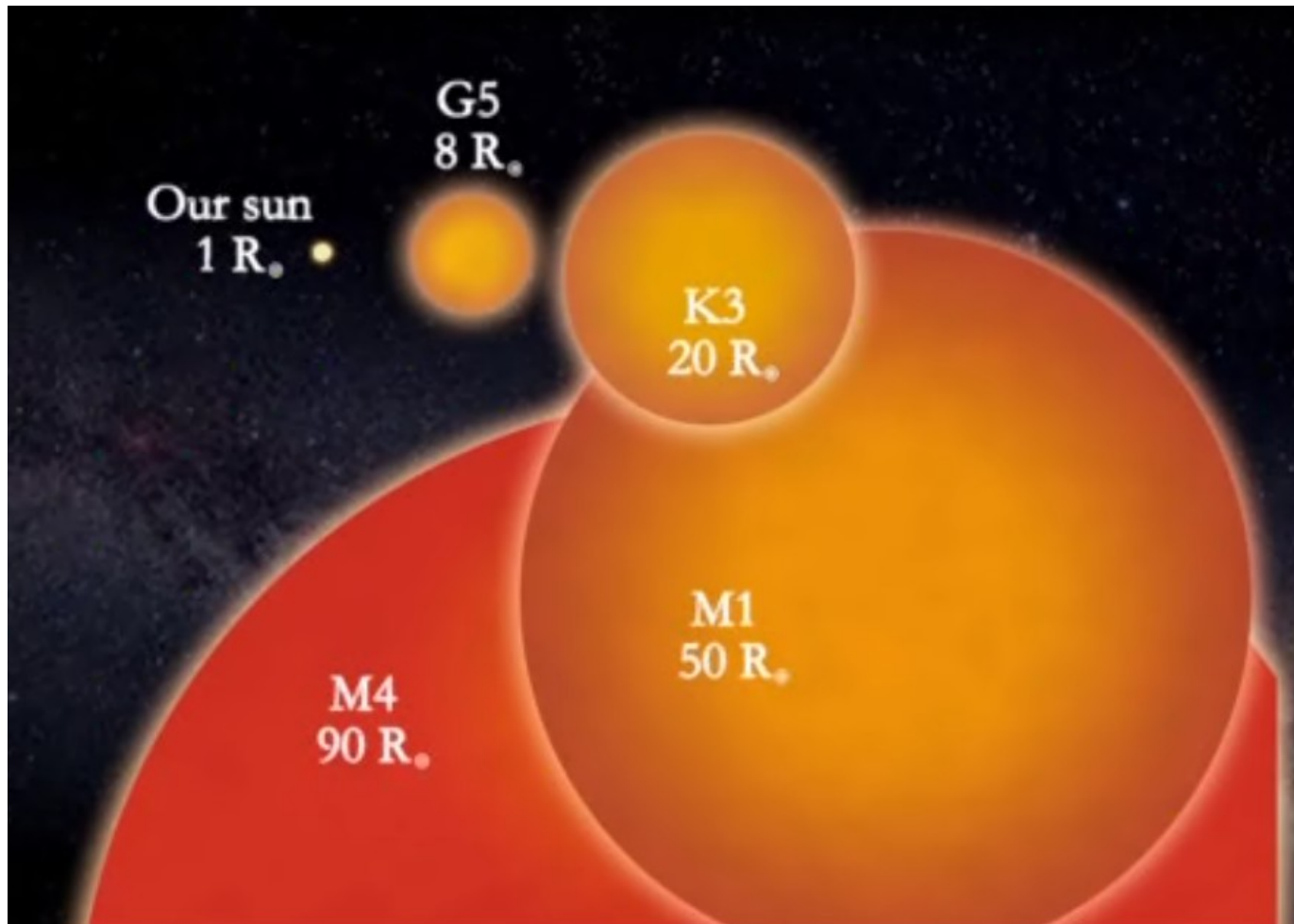


The luminosity is 1 million times the luminosity of the sun. So the size is x 1000 (same temp).
The size is 4.6 AU (picture by Hubble space) $\rightarrow 1000^2 = 1000,000$



The radius (and thus surface area) of **Betelgeuse** must be larger to output the same power as **Rigel**!

In fact, **Betelgeuse** is 1000x the size of the Sun, while **Rigel** is merely 62x the size of the Sun. Both have luminosities that are $\sim 100,000x$ that of the sun!

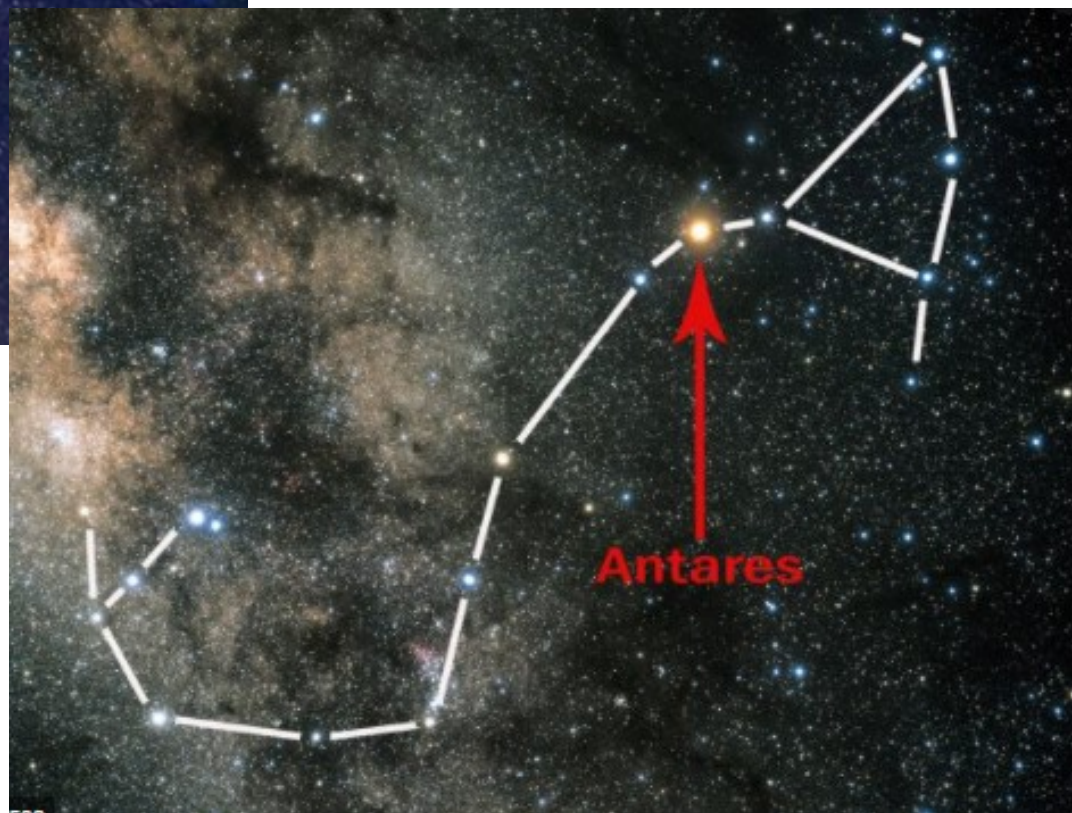


Luminosity also depends on the size of stars too.
Stars can have low temperature but big size so luminous



southern hemisphere of the sky

It's classified as an M1 red supergiant star. The M1 designation means that Antares is reddish in color and much cooler than many other stars. Its surface temperature is about 6,100 degrees F (3,400 degrees C).

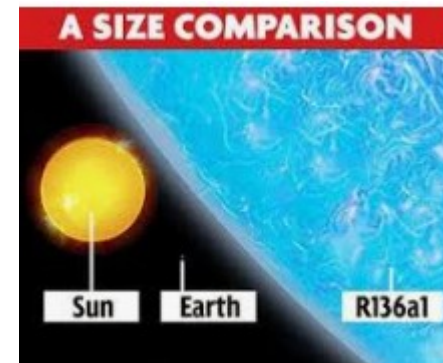


cR136a1 is located in the R136 star cluster, which is located in the Large Magellanic Cloud – a galaxy satellite of the Milky Way. The star gained its title as the most massive and luminous star ever discovered in 2010

163,000 light years
away.

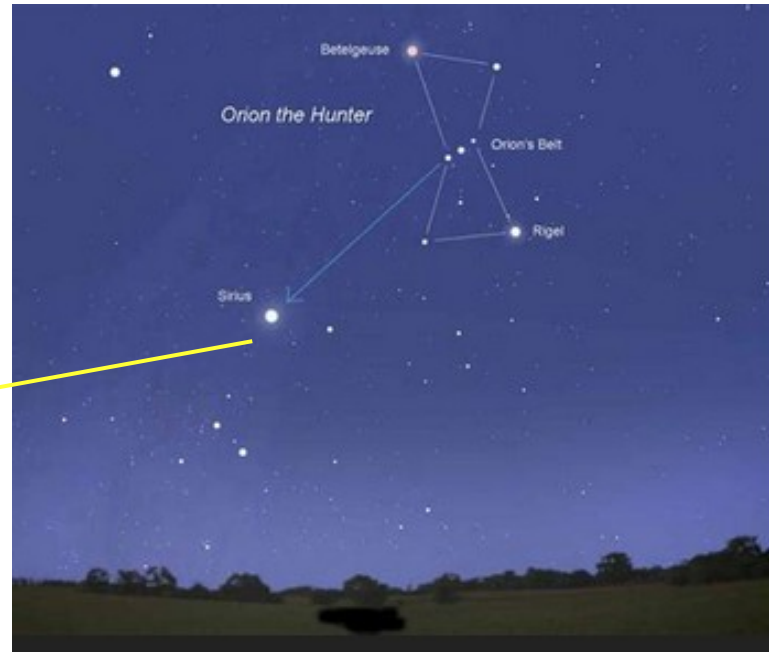
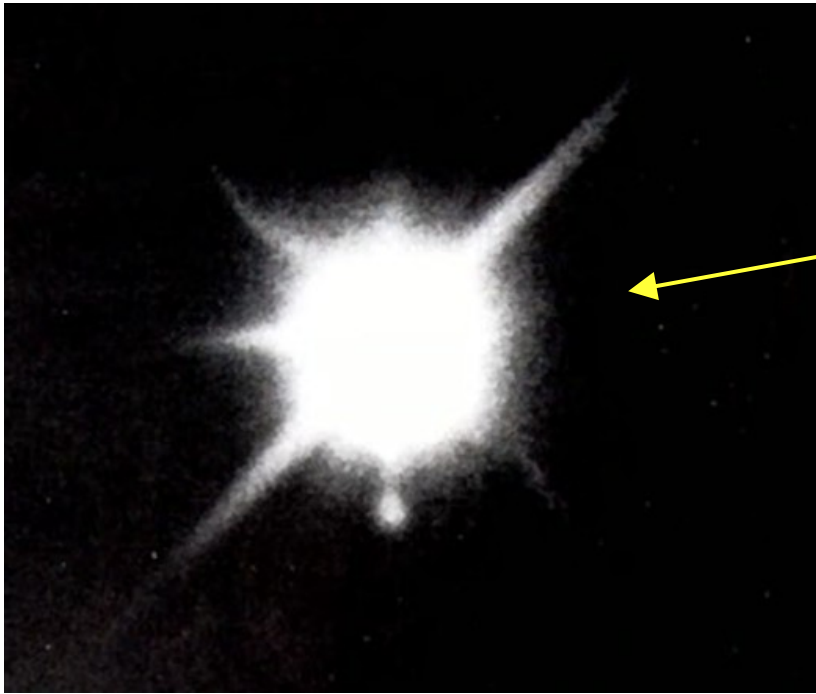


the star R136a1 has just shattered all of the records.

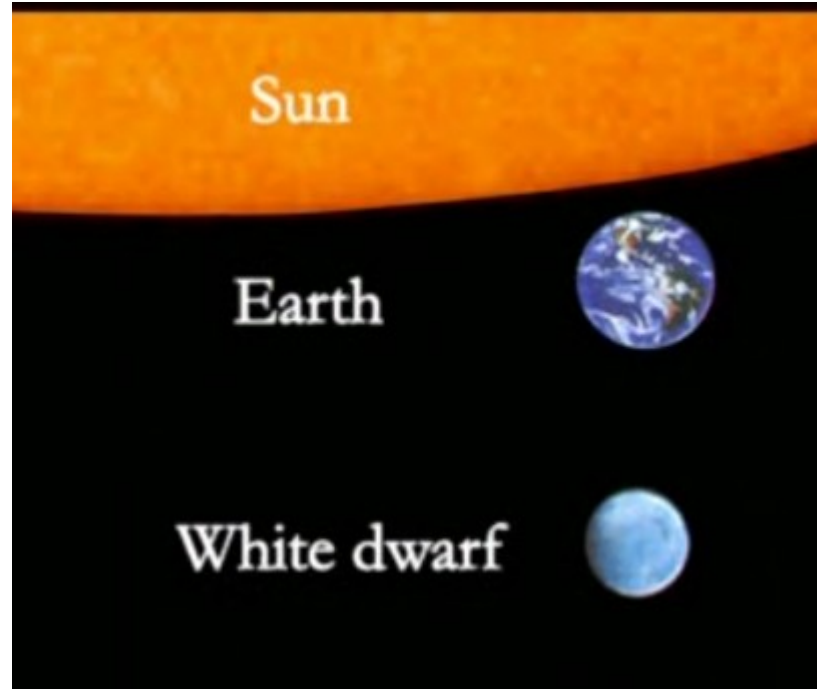


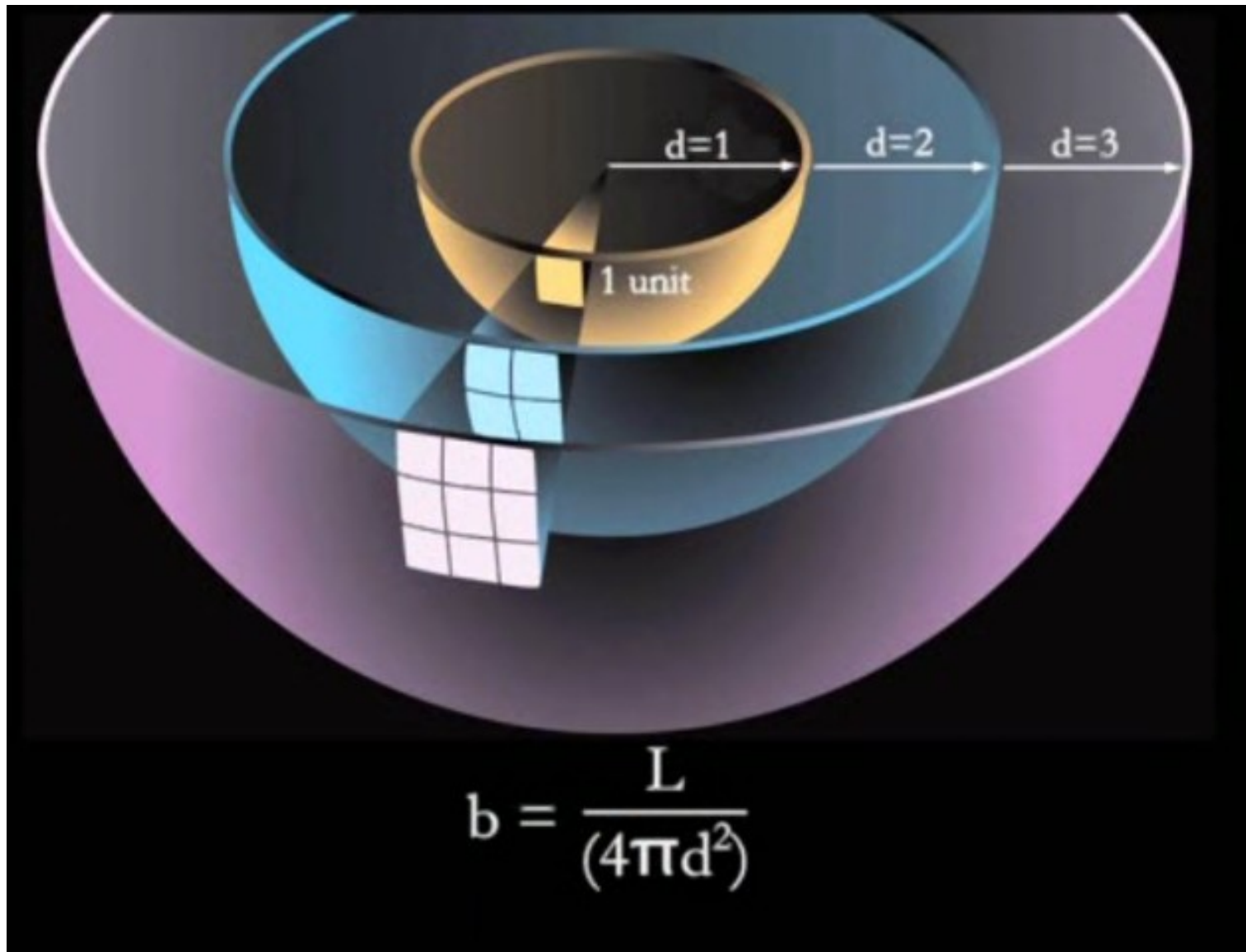
It weighs in at 265 times the mass of the Sun. It's so massive that it's probably already blown off something like 60 times the mass of the Sun, meaning it was over 300 times as massive as the Sun when it was born. With a lifetime under 10,000 years.

Movie scale



The companion of Sirius A
Is a white dwarf.
For the same temp than the Sun, its
Less luminous. So smaller.
10,000 less luminous →
100 times smaller ($100 \times 100 = 10,000$).





Don't confuse R and d.

R is the size of the star and d is the distance between the observer and the star.



Some stars have variable luminosity like Mira.



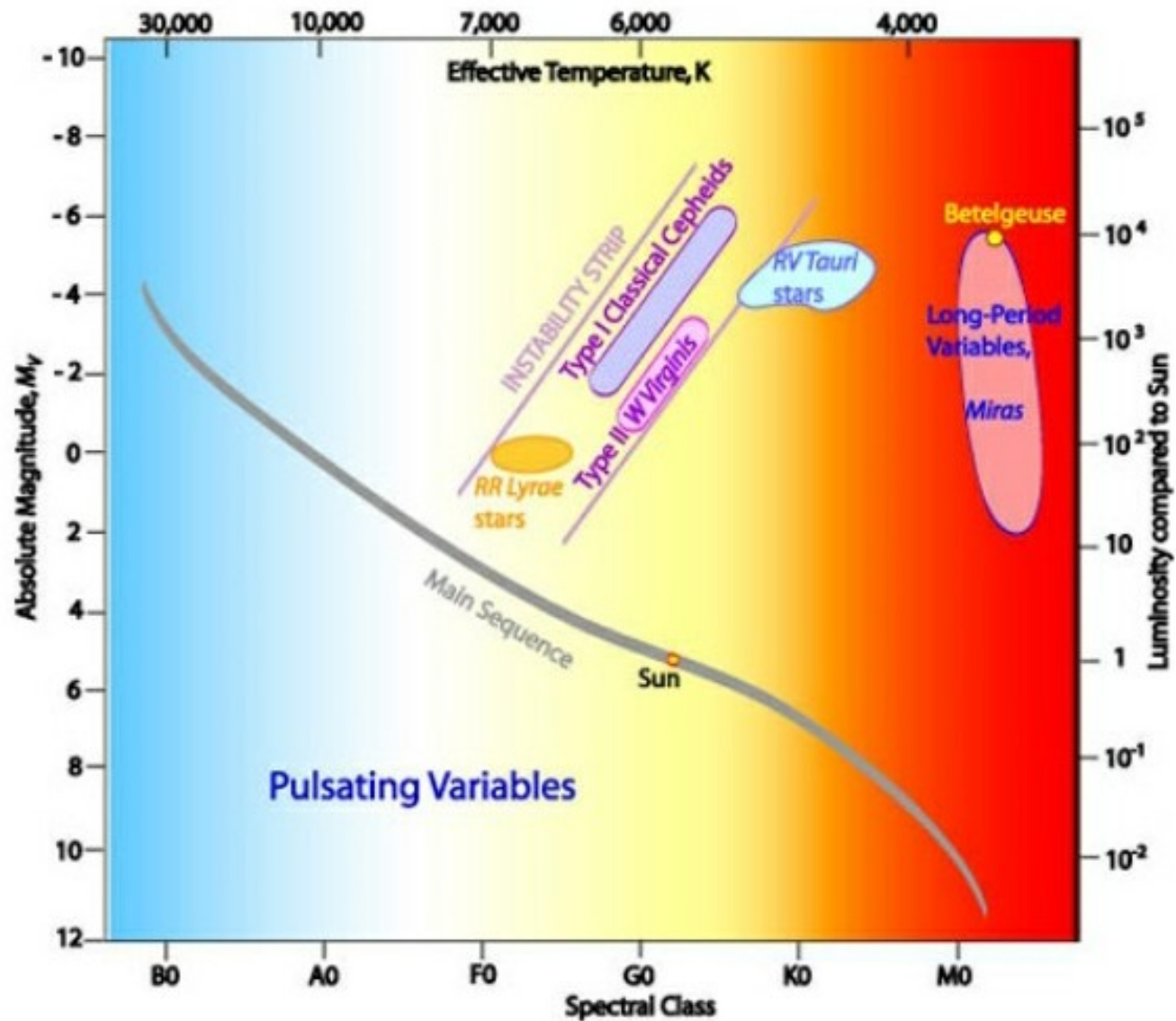
One of the most famous examples of this type is the long-period variable star Mira

Mira is a variable star that undergoes periodic changes in brightness every 332 days, ranging from a maximum brightness of around 3.5 visual magnitudes to a minimum brightness of about 9 magnitudes.

— the first variable star to be discovered. Mira is located in the constellation Cetus, known by the ancient Greeks as the monster that was about to attack Andromeda when the hero Perseus arrived just in time to destroy it.

The giant has the stellar classification M7 IIIe and its spectral class varies from M5 IIIe to M9 IIIe as it pulsates.

Movie variable



The location of types of pulsating variables on the HR diagram.

Very important to measure distance → cepheid variable stars

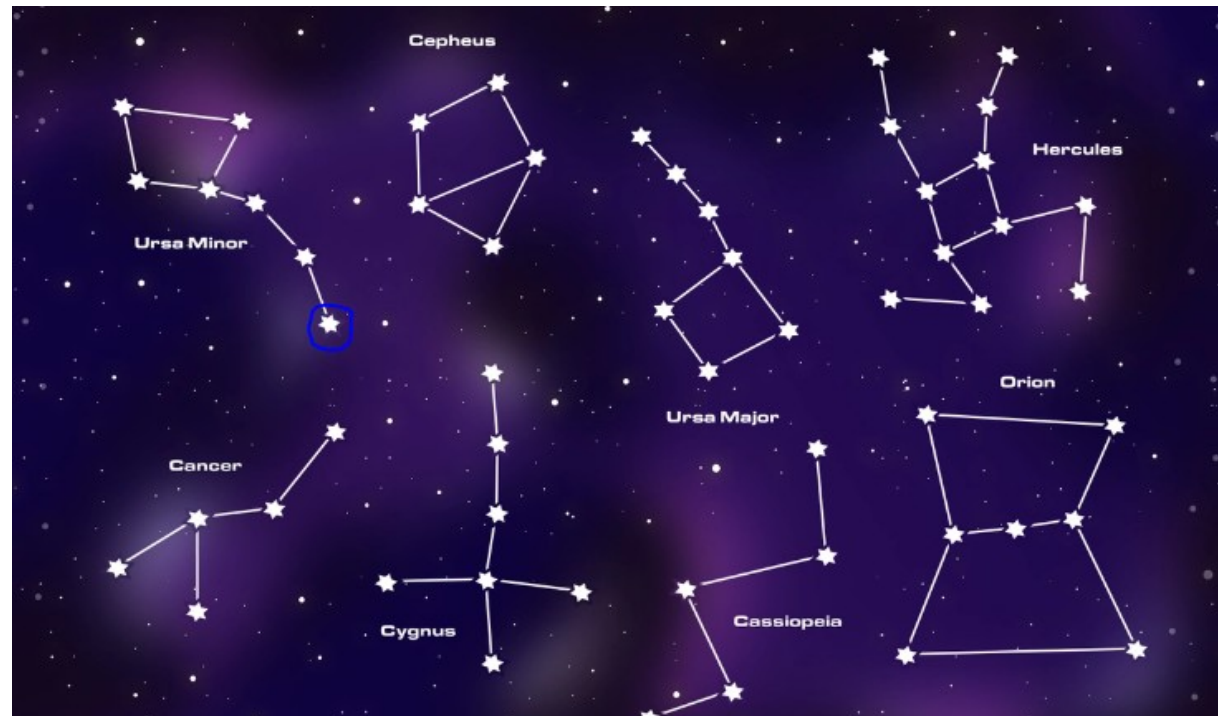
The pattern was first noticed in 1784 in the constellation Cepheus (“king”) in the northern sky, so these stars became known as “Cepheid variables.” Cepheid variables went from interesting to completely indispensable in the early 1900s thanks to the work by astronomer Henrietta Leavitt.

Edward Pigott detected the variability of Eta Aquilae, the first known representative of the class of classical Cepheid variables. The eponymous star for classical Cepheids, Delta Cephei, was discovered to be variable by John Goodricke a few months later.

Delta Cephei (δ Cep, δ Cephei) is the Bayer designation for a quadruple star system located approximately 887 light-years away in the northern circumpolar constellation of Cepheus, the King.

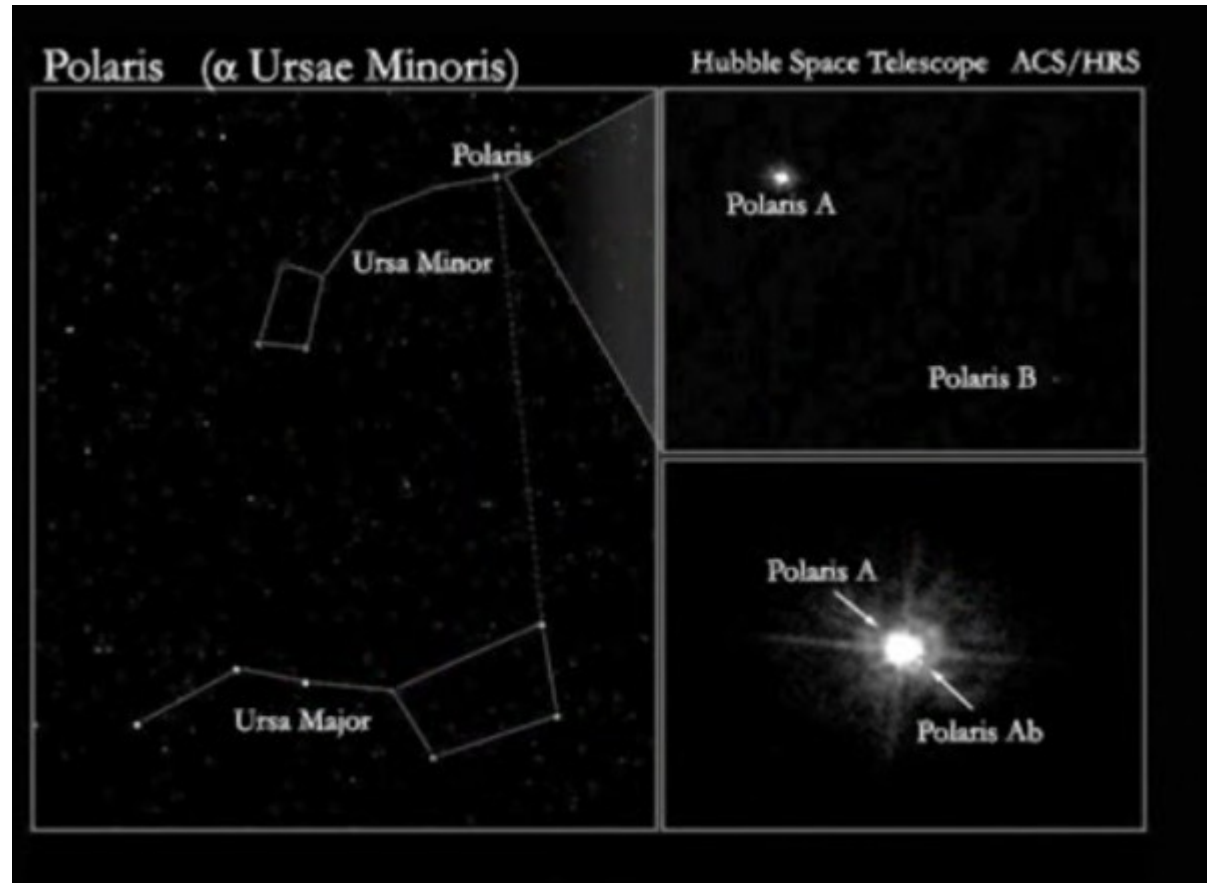


best seen during culmination
during the month of November



Very important to measure distance → cepheid variable stars

Although appearing to the naked eye as a single point of light, Polaris is a triple star system, composed of the primary, a yellow supergiant designated Polaris Aa, in orbit with a smaller companion, Polaris Ab; the pair is in a wider orbit with Polaris B





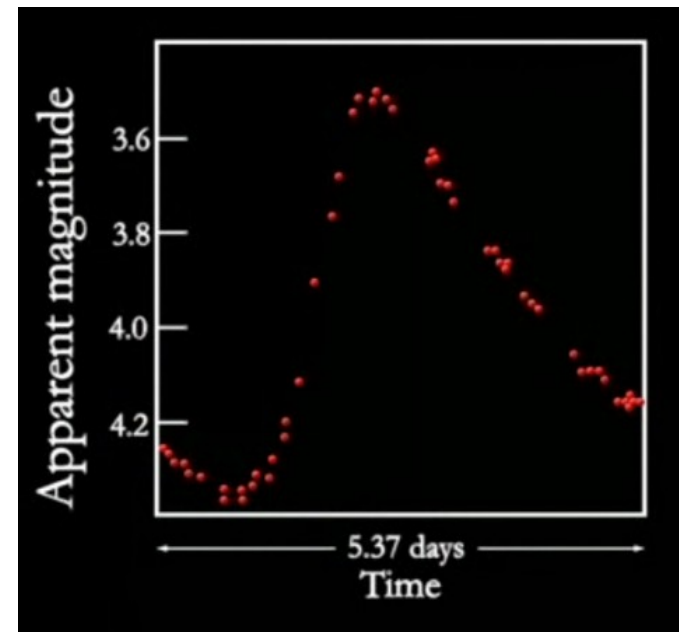
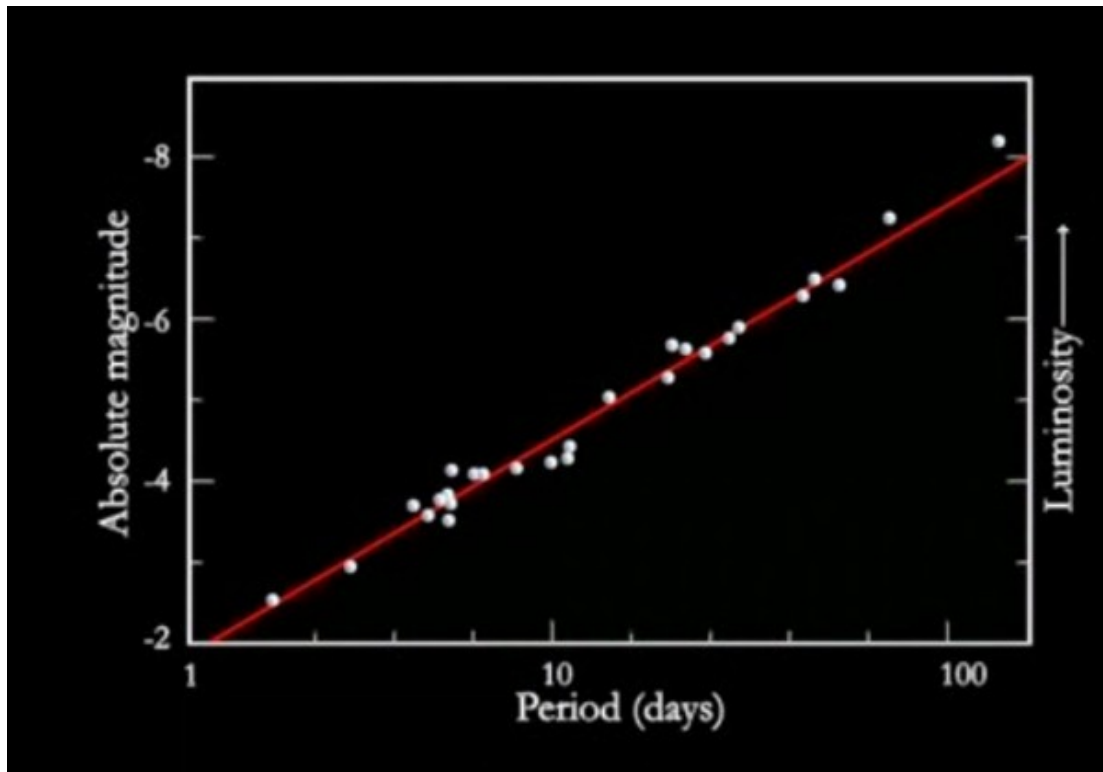
Shortly after her graduation from what we now call Radcliffe, an illness caused Henrietta Swan Leavitt to lose her hearing. The Harvard College Observatory eventually hired her as a human “computer.” Her job: review the hordes of glass photographic plates and calculate the brightness of the stars in them. She discovered **Cepheid variables**.

All the stars in the Magellanic clouds are at the same distance.
They have Cepheid Variables.

Their luminosity is a function of their period.

→ use them as standard candles.

Cepheids are extremely important to astronomers because their periodic changes in brightness can be used to figure out how far away they are. Because Cepheids are intrinsically bright stars, they can be seen far away, and astronomers can use them to determine the distance to very distant galaxies.

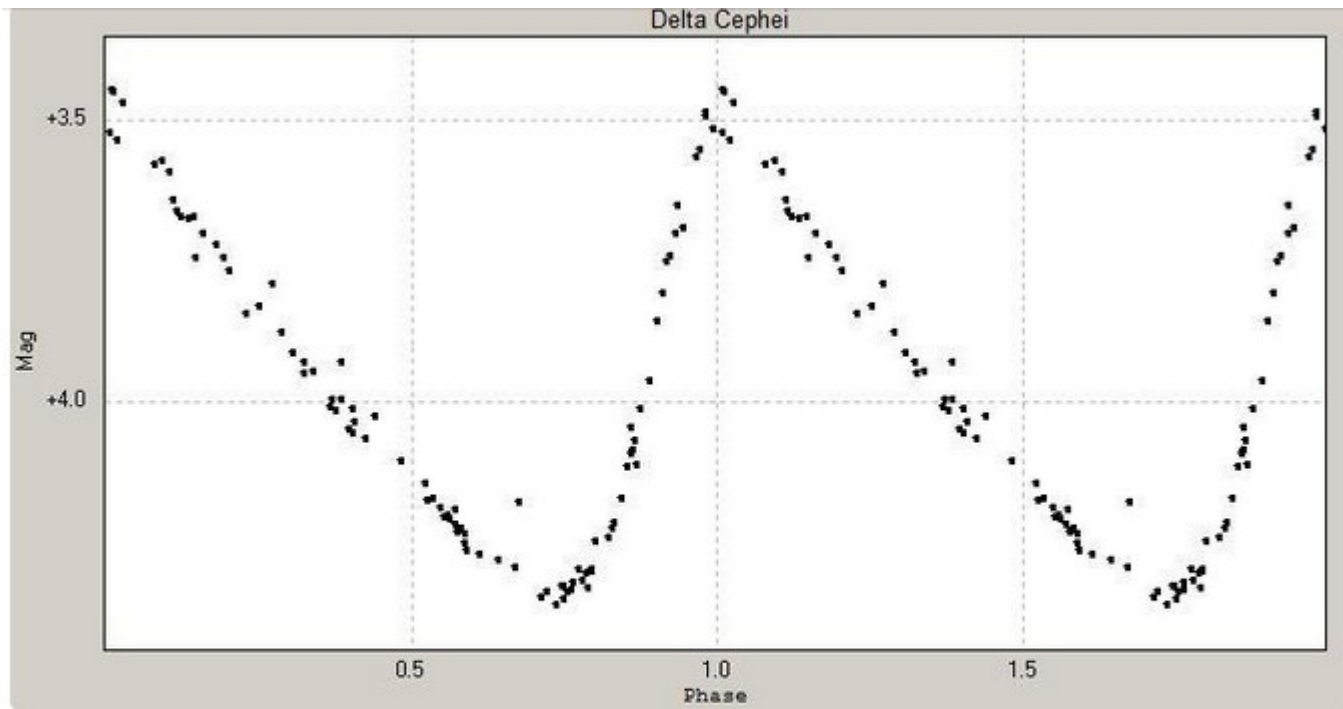


Movie about henrietta Leavitt

170 l.y. away. Large cloud

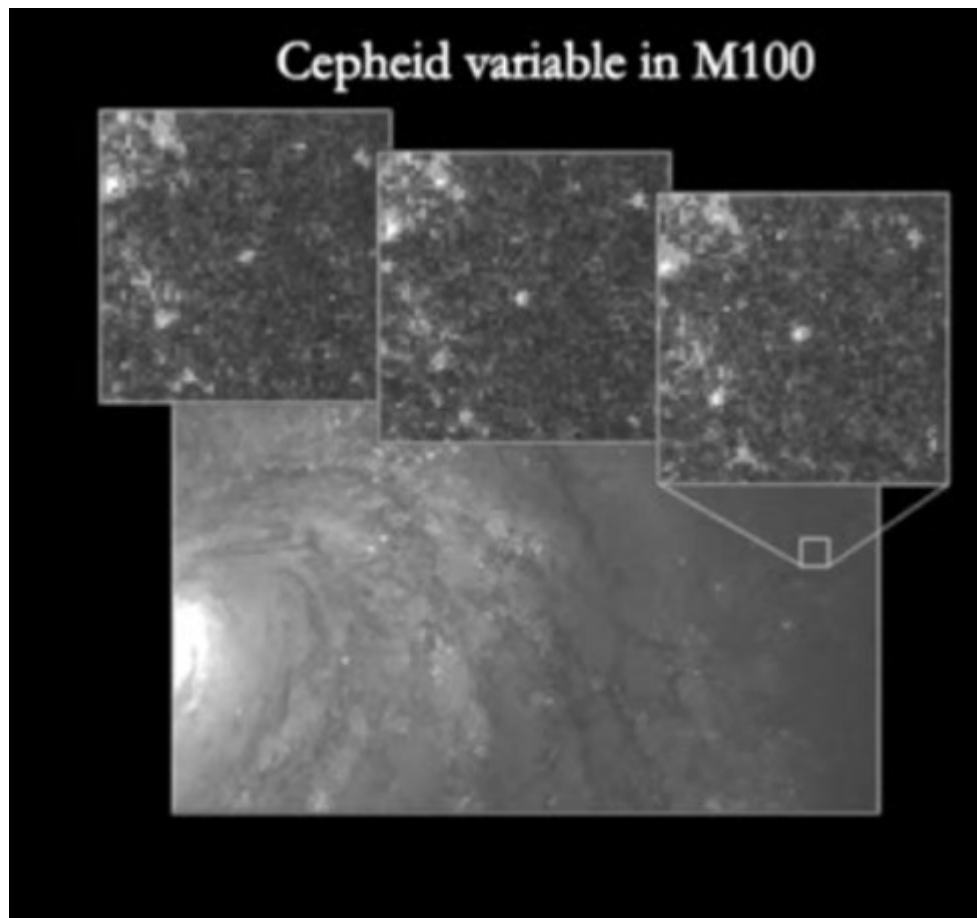


Delta Cephei displays about a two-fold change in brightness (0.23 visual magnitudes) every 5.366 days, ranging from a visual magnitude of 3.48 at its brightest to 4.37 at its faintest. Zeta and Epsilon Cephei are useful comparison stars for noting changes in brightness of Delta Cephei from one night to the next.



A light curve plot, brightness vs. time, of the changes in brightness in Delta Cephei. At the bottom, the two lowest sections are when the star is at its minimum brightness. The time it takes from one minimum to the other is 5.366 days. The y-axis shows the star's brightness, in units of magnitude. Image via ThomasK Vbg /

<https://earthsky.org/brightest-stars/delta-cephei-the-kings-famous-variable-star/>



Messier 100 (NGC 4321) is a spiral galaxy located in the constellation Coma Berenices in the Virgo Cluster of galaxies. M100 is 55,000,000 light years away from Earth